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Novel Dry-Powder Pickering Emulsion Approach for Efficient Hydrogen Peroxide Encapsulation and Polymer Schmoo Degradation

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Abstract

A simple single-step method to microencapsulate an aqueous hydrogen peroxide (H_2O_2) solution and form a water-in-air emulsion is presented based on the Pickering emulsion technology to address handling, transport, and controlled release of hydrogen peroxide in oil fields. The encapsulation was performed by mixing a 5-weight percent H_2O_2 solution with hydrophobic silica nanoparticles in a highspeed blender. Droplet morphology was examined using optical microscopy. The efficacy of retarding corrosion by microencapsulating H_2O_2 was performed on mild steel sample blocks by following the specific ASTM standard. The release of encapsulated H_2O_2 droplets was visibly observed by mixing a small amount of coloring dye. It was quantified by conductivity measurements owing to the presence of salt in the aqueous solution of H_2O_2 prior to encapsulation. Finally, field polymer deposit samples were subjected to degradation tests by immersing them in the H_2O_2 solution. The silica nanoparticles synthesized via the sol-gel method exhibited an average primary particle diameter of approximately 38.59 nm. Microscopic analysis indicated that the Pickering emulsion droplet sizes were in the order of a few microns. An addition of a small amount of hydrophilic surfactant effectively changed the wetting nature of the hydrophobic silica nanoparticles, consequently breaking the emulsion as reflected by the conductivity measurement results. Corrosion inhibition results suggest significant corrosion mitigation. Additionally, field-deposited polymers successfully degraded upon treatment with H_2O_2 , validating its practical utility.

Introduction

Pickering emulsions refer to emulsions stabilized by solid particles rather than conventional surfactants. One of the important applications of Pickering emulsion for encapsulation came about when Degussa (Now under Evonik Industries) patented a water-in-air type emulsion later marketed as dry water (Schutte et al., 1965). The dry emulsion was stabilized using hydrophobic particles, and the overall matter contained more than 90 percent by weight of water. The emulsion showed high stability under ordinary conditions and showed flow properties akin to a powdered solid. A seminal paper (Aussillous & Quere, 2001) reported the flow characteristics of a single droplet of these powdered emulsions (referred to as liquid marble) on an

inclined surface, highlighting minimal frictional losses, rendering them a great source of transport of small liquid volumes encapsulated within a non-wetting shell. Forny in multiple works (Forny et al., 2007, 2009, 2010) reported the formation of dry water using two different techniques, namely, atomization and High-speed shear mixing, with the former resulting in smaller units compared to the latter. They also reported the contact angle measurements relating to the formation of mousse and the minimum value for a stable dry emulsion. Using the same principles, successful encapsulation of a ferrofluid liquid in a layer of hydrophobic silica particles as a dry ferrofluid-in-air emulsion while retaining all the original ferrofluidic properties has also been reported (Sousa et al., 2013). Finally, a study for acid fracturing (Singh et al., 2017) reported encapsulation of hydrochloric acid as dry powder for etching of shale rock.

Polymer flooding has been a vital chemical enhanced oil recovery method used as early as the mid-1960s, owing to increased sweep efficiencies (Sandiford, 1964). The polymer of choice for the flooding process in almost all major fields around the world has been the partially hydrolyzed polyacrylamide (HPAM) owing to its desirable properties (Jain et al., 2023) and low-cost compared to other polymers (Seright & Wang, 2023). One of the significant issues with HPAM flooding is that few studies have reported fouling of field equipment, possibly due to interactions with multivalent ions in the presence of dissolved oxygen (Mittal et al., 2018). Hydrogen peroxide may be used to address these fouling challenges as it can break down the three-dimensional cross-linked polymer lumps. However, handling, transportation, and injection of hydrogen peroxide solution is challenging as it is corrosive in nature and may damage pipelines and equipment. Thus, to address this corrosion challenge, we present this study on the encapsulation of hydrogen peroxide as dry powdered droplets encapsulated within hydrophobic silica nanoparticles. This work is presented with an objective of extreme reduction in corrosion rates, consequently addressing the problem of downtimes while at the same time presenting a methodology for the release of hydrogen peroxide at the point of interest, thus avoiding direct contact of hydrogen peroxide with materials prone to corrosion in the oil field. A single-step, straightforward method of preparation followed by characterization is discussed. Finally, a simple process of breaking down these emulsions by using a hydrophilic surfactant solution is presented. The overall objective is backed up by strong corrosion inhibition data.

Materials and Methods

Materials

Hydrogen peroxide (30% by weight, Emplura grade), Trichloro(octadecyl)silane ($\geq 91\%$), Tetraethyl orthosilicate (reagent grade, 98%), and C₁₂ α -olefin sulfonate (AOS) were purchased from Merck and obtained from Pilot chemical company, respectively. Ethanol (premium grade, 100%) was purchased from Hayman. Salts such as sodium chloride (99.5%), ammonia solution (25%), and solvents such as methanol (99.5%), acetone (99%), and toluene (99%) were purchased from Loba Chemie Pvt. Ltd. Mild steel sample blocks were used for corrosion inhibition tests. All chemicals were used as received. Deionized (DI) water with a conductivity of 18 gS/cm at 20°C was used for dilution and rinsing.

Methods

Synthesis of Hydrophilic Silica Particles. Powdered hydrophilic silica nanoparticles were prepared based on the well-known sol-gel method. 5 mL of tetraethyl orthosilicate, along with 30 mL of ethanol, were taken in a reagent bottle and sonicated at a low frequency of 42 KHz in a water bath sonicator for 10 min. 1 mL of DI water was added to the solution at a rate of 0.2 mL/min for hydrolysis of the silica precursor, i.e., tetraethyl orthosilicate, and the solution was sonicated for 90 min. Then, 2 mL of ammonia solution (25%) was added at a rate of 0.2 mL/min, and the solution was again left for sonication for 180 min. The solution, covered with a paraffin film, was left overnight for gel formation. The solution was washed with ethanol via centrifugation three times at 8000 rpm for 8 min and washed with DI water until a pH close to neutral

was achieved. The supernatant was discarded after each wash, and the final sedimented solid was left for drying at 80°C in an oven for 24 hr. Finally, the sedimented solid was calcined at 600°C for 2 hr.

Hydrophobization of Silica Particles. Hydrophilic silica nanoparticles were mixed with a solution of 7% trichloro(octadecyl)silane in toluene and left for sonication for 2 hr at 80°C. The mixture was kept at 80°C for 7 days in an oven. Next, the solution was centrifuged at 8000 rpm for 20 min, and the supernatant was discarded. The silica nanoparticles were then washed with toluene, acetone, and methanol, in that order, by centrifugation at 8000 rpm for 8 min to eliminate any remaining polar or non-polar impurities with the supernatant discarded at each step. Subsequently, the particles were left for 24 hr in an oven at 80°C for drying.

Characterization of Silica Particles. A FEI Titan G2 60-300 scanning transmission electron microscope was used to analyze the morphology of hydrophilic silica nanoparticles. A small amount of powdered sample was sonicated with ethanol for 1 hr in a bath-sonicator and then deposited on a gold-coated TEM grid and dried. The dried grid was then analyzed via TEM imaging. ImageJ software was used to analyze particle morphology.

Preparation and characterization of dry powdered emulsion of hydrogen peroxide. Hydrogen peroxide (30 wt%) solution was diluted down to 5 wt% using deionized water. Hydrophobic silica nanoparticles and diluted peroxide solutions were mixed in varying ratios of 1:5, 1:10, and 1:15, respectively, in a high-speed, indigenously made homogenizer at a speed of 24000 rpm for 40 sec. This resulted in the formation of the dry water-in-air powdered emulsion, with water containing 5 wt% H₂O₂ and silica nanoparticles completely encapsulating the liquid. The mixture was sheared in a 50 mL conical bottom polypropylene tube of 27 mm internal diameter. The prepared water-in-air emulsion was stored in the same conical bottles and left for further investigation. The prepared emulsions of three different ratios were analyzed with a Leica DM 2700M polarization microscope under incident light to study the droplet size distribution and morphology of the three samples.

Corrosion inhibition tests. A total of six samples of mild steel blocks, after cleaning with acetone and subsequently with DI water, were measured for their original weight and kept in two different samples, namely a 1:10 ratio powdered emulsion and a 5 wt% hydrogen peroxide solution, for a period of approximately ten days under ambient conditions in a sealed environment. The total amount of 5 wt% hydrogen peroxide was kept constant at 10 mL in both cases. The final weights of the samples were measured at the end of ten days, and the corrosion rate was reported corresponding to the literature (Salarvand et al., 2017). Corrosion rates were calculated based on the ASTM standard (Guide for Laboratory Immersion Corrosion Testing of Metals, 2004) as shown in Equation 1.

$$C_{\text{rate}} = \frac{K * W}{A * t * \rho} \quad (1)$$

Where:

C_{rate} = corrosion rate in mm/year

K = 8.76 × 10⁴ (conversion constant)

W = loss in mass in grams

A = initial exposed surface area in cm²

t = immersion time in hr

ρ = density of mild steel, 7.85 g/cm³

Release of encapsulated hydrogen peroxide solution. A qualitative and quantitative analysis of the release of peroxide solution from the water-in-air emulsions under the action of surfactant solution was studied. The water-in-air emulsions encapsulating 5 wt% H₂O₂ and a small amount of coloring dye were prepared. The prepared emulsions were first submerged in DI water. The emulsions were found to be stable in the DI

water. A known amount of AOS surfactant solution was added to these emulsions to trigger the release of the encapsulated aqueous solution containing H_2O_2 and the dye. The sample images were taken at regular time intervals to study the release of encapsulated aqueous solution based on the color change of the solution. The release was quantitatively obtained by measuring the increase in the conductivity of the deionized water, containing water-in-air emulsions, when adding the hydrophilic surfactant. In this test, 0.5 wt% NaCl was also added to the encapsulated aqueous solution of 5 wt% H_2O_2 prior to encapsulation. To carry out these tests, a given amount of encapsulated powder (water-in-air emulsions) was placed in two tubes containing DI water. In one of these tubes, a known amount of AOS solution was added, whereas the other sample was left as is. The conductivity of both samples was measured continuously and compared to gain qualitative insights into the release process on adding the hydrophilic surfactant.

Polymer degradation by hydrogen peroxide. The breakdown of polymer lump samples, collected from the Mangala field, using 5 wt% H_2O_2 , was studied. The polymer lump samples were kept in the aqueous solution of 5 wt% H_2O_2 and visually observed for degradation at room temperature. For comparison, the polymer lump samples were also kept in DI water.

Results

Synthesis and characterization of Silica Particles

Hydrophilic silica nanoparticles were prepared by the typical sol-gel approach (Keane et al., 2010; Stober et al., 1968), which involves hydrolysis of the silica precursor, namely tetraethyl orthosilicate, followed by condensation of ethoxysilanol with tetraethyl orthosilicate or another silanol with ammonia acting as a morphological catalyst. The hydrophobization of the nanoparticles was carried out via reaction with silanes (Wei et al., 1993).

TEM imaging of the hydrophilic nanoparticles was performed to study the morphology and size of individual particles formed. Fig. 1 shows spherical morphology with a primary particle of average diameter of 38.59 nm with a standard deviation of 4.81 nm.

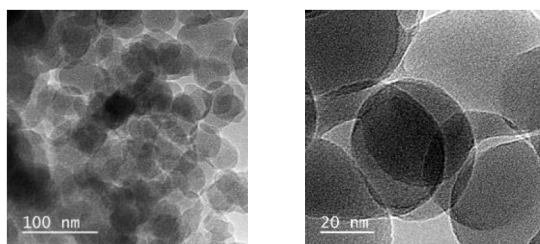


Figure 1—Transmission electron microscopic image of silica nanoparticles prepared at different magnifications

Preparation and characterization of dry powdered emulsion of Hydrogen peroxide

The water-in-air emulsions, with 5wt% H_2O_2 encapsulated, were prepared by shearing a fixed ratio of hydrogen peroxide solution and hydrophobic silica nanoparticles, as shown in Fig. 2. Fig. 2(a) shows the components before mixing, while Fig. 2(b) shows the formation of dry powder-like emulsions with the aqueous solution completely encapsulated with the framework of silica nanoparticles. Fig. 2(c) represents the flowing nature of the prepared emulsion through a funnel akin to conventional powder-like materials.



Figure 2—Method of preparation of water-in-air powdered emulsion (a) before mixing; (b) after mixing; (c) the free-flowing nature of the final product through a funnel

Optical microscopic image analysis of the prepared emulsions in the encapsulation ratios of 1:5, 1:10, and 1:15 (encapsulation ratio is defined here as the ratio of mass of silica nanoparticles to the mass of aqueous solution) was carried out to analyze the morphology and droplet sizes of the formed emulsion. Fig. 3 shows the microscopic images of these emulsions of varying encapsulation ratios under incident light using a green filter for enhanced contrast. The images clearly show hydrogen peroxide solutions encapsulated as micronized stable droplets, with the average droplet diameter increasing in direct correlation with the amount of encapsulation. The images were analyzed using ImageJ software, and the average droplet diameter for the corresponding encapsulation ratios was calculated to be 1.90, 4.77, and 5.45 μm for the ratios 1:5, 1:10, and 1:15, respectively.

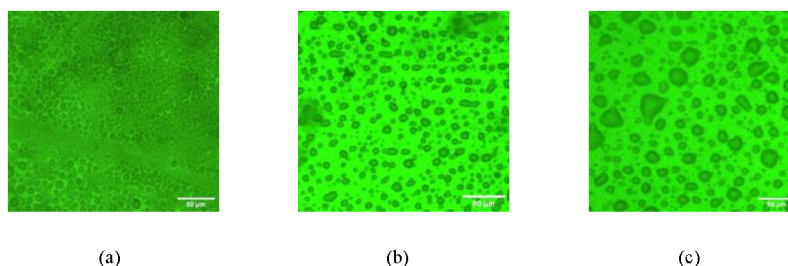


Figure 3—Microscopic images of the prepared emulsions corresponding to the encapsulation ratios of (a) 1:5, (b) 1:10, (c) 1:15. Bar scale corresponds to 50 μm in each case

Release of encapsulated peroxide solution

Fig. 4 shows a time lapse of the solution kept in contact with the surfactant solution. A change in intensity of the muddy green color taking over the solution texture clearly shows the breakdown of the emulsion on the addition of AOS. Fig. 5 shows conductivity readings with time for the two samples. It is evident from the figure that without the presence of surfactants in the solution, the emulsion remained intact with a minor increase in conductivity attributed to the diffusion of salt into the bulk phase. The conductivity value of over 13300 gS/cm measured around a time of approximately one hour implies almost complete breakdown of the emulsion under the action of AOS.



Figure 4—Visual inspection of the release of peroxide solution under surfactant action

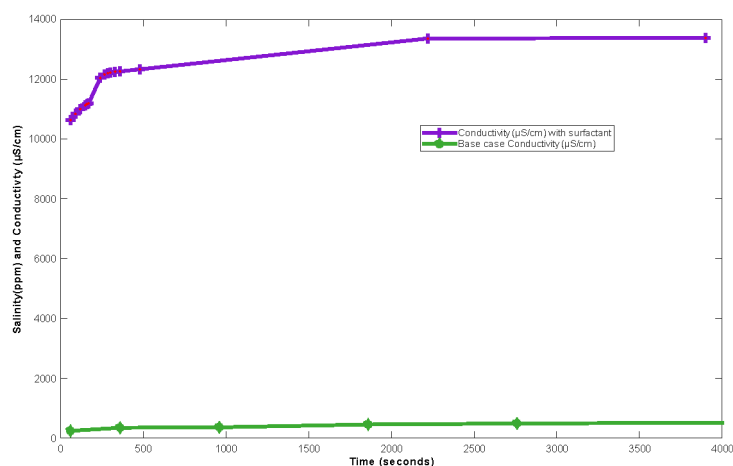


Figure 5—Conductivity of solution as a function of time with and without AOS surfactant

Corrosion inhibition tests

Corrosion inhibition efficiency tests were conducted on mild steel samples, which were exposed for approximately ten days. Six samples, three for each case, namely: (i) 5 wt% hydrogen peroxide solution, and (ii) encapsulated 5 wt% H_2O_2 solution. Corrosion rates were determined gravimetrically, utilizing a weighing balance with a precision of 0.001 g. Table 1 shows the corrosion rates for all three samples for both cases. Samples immersed in the solution 5 wt% H_2O_2 without any encapsulation exhibited an average corrosion rate of approximately 0.35 mm/year, indicating aggressive corrosive damage. Remarkably, the encapsulated H_2O_2 solution demonstrated complete corrosion inhibition, yielding zero measurable corrosion over the test period.

Table 1—Corrosion rates on mild steel samples for non-encapsulated and encapsulated 5% H_2O_2 solution

Sample	Specimen Number	Corrosion rate (mm/year)
Base Case (Non-encapsulated H_2O_2 solution)	1	0.4190
	2	0.3402
	3	0.2965
Water-in-air type powdered emulsion	4	0
	5	0
	6	0

This substantial reduction in corrosion through dry encapsulation of H_2O_2 solution offers significant implications for equipment longevity, operational efficiency, and maintenance cost reduction in oil field operations.

Polymer degradation by hydrogen peroxide. Fig. 6 represents the initial and final snapshots of the polymer lump kept under the action of non-encapsulated hydrogen peroxide solution over a period of five days. Clearly, as already reported in the literature, the hydrogen peroxide solution is extremely useful in degrading the lump. Fig. 6(a) represents the initial snapshot, Fig. 6(b) is the side view, and Fig. 6(c) represents the top view showing the degraded state of the lump into smaller fragments.

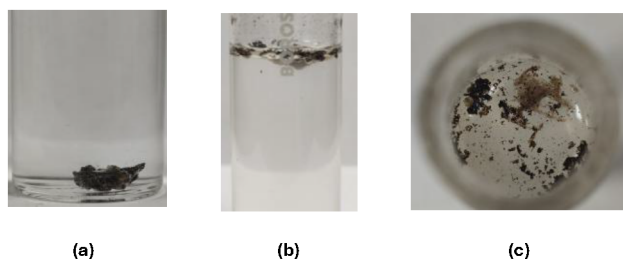


Figure 6—Effect of 5 wt% solution of hydrogen peroxide on polymer lump (a) initial image of lump in peroxide solution, (b) side view and (c) top view of same sample after a period of five days

Conversely, the procedure was repeated with DI water as the base case. Fig. 7(a) represents the initial snapshot of the polymer lump in DI water, and it can be clearly seen from Fig. 7(b) that the lump remains intact over an extended period of time with no degradation into smaller fragments, apart from some swelling effects.

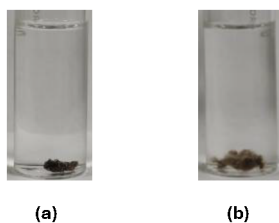


Figure 7—Polymer lump in deionized water (a) initial image of lump, (b) sample after a period of five days

Conclusions

A single-step method to encapsulate a 5 wt% solution of hydrogen peroxide was demonstrated in this paper with the primary objective of preventing field allied equipment from corrosion damage and enhancing the overall efficiency of the oil recovery process. Silica nanoparticles, hydrophilic in nature, were synthesized by the simple hydrolysis and condensation of silica precursor with ammonia as morphological catalyst, which were then hydrophobized by reaction with trichloro(octadecyl)silane. The encapsulation was performed by shearing the mixture of the peroxide solution and hydrophobized particles in various ratios in a high-speed blender. The following are the main conclusions summarized below:

- i. The prepared silica nanoparticles by the sol-gel method exhibit a primary particle diameter of 38.59 nm.
- ii. 5 wt% hydrogen peroxide solution was easily encapsulated within a framework of hydrophobic silica nanoparticles by single-step shearing at high rpm.
- iii. Microscopic analysis of the formed emulsions revealed stable encapsulated droplets with sizes of the order of a few microns.
- iv. The breakdown of emulsion under the action of 1% AOS solution resulted in almost complete release of the encapsulated media in just over an hour under ambient conditions.
- v. The corrosion inhibition effectiveness of these water-in-air emulsions was backed up by a zero measurable corrosion rate for the specimen in contact with the water-in-air emulsion system under the reported testing conditions.
- vi. The entire exercise was backed up by the data for degradation of a real field polymer by hydrogen peroxide solution.

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